



RIVER VALLEY HIGH SCHOOL
2020 JC2 Preliminary Examination
Higher 2

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CLASS					
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INDEX NUMBER		
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MATHEMATICS

9758/02

Paper 2

21 September 2020

3 hours

Candidates answer on the Question Paper
 Additional Materials: List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Write your answers in the spaces provided in the question paper.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved graphing calculator is expected, where appropriate.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

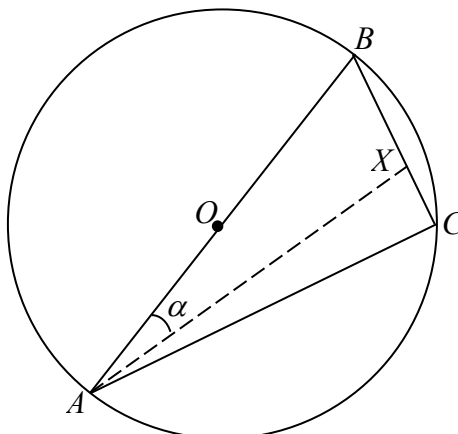
The total number of marks for this paper is 100.

For examiner's use only	
Question number	Mark
Total	

Calculator Model:

Section A: Pure Mathematics [40 marks]

1.



The diagram above shows a circle with centre O . The diameter AB has length 13 units and the point C on the circumference of the circle is such that AC is 12 units. X is a point on BC such that angle BAX is α radians.

(i) Show that $\tan \angle BAC = \frac{5}{12}$. [1]

(ii) By considering triangle XAC , or otherwise, show that $CX = \frac{60 - 144 \tan \alpha}{12 + 5 \tan \alpha}$. [3]

(iii) It is given that α is sufficiently small for α^3 and higher powers to be ignored, show that $CX \approx 5 + p\alpha + q\alpha^2$, where p and q are exact constants to be determined. [3]

(iv) Explain why $\alpha = 0.6$ is not suitable to be used to find CX in part (iii). [1]

2. A particle moving in a liquid is such that after t seconds, its velocity is $v \text{ ms}^{-1}$, $v \neq 0$.

v satisfies the differential equation $v \frac{dv}{dt} + 2v^2 = te^{-2t}$.

(i) By using the substitution $y = ve^{2t}$, show that $y \frac{dy}{dt} = te^{2t}$. [3]

(ii) Hence find v^2 in terms of t where $\frac{dv}{dt} = -8$ when $t = 0$. [6]

3. A function f is said to be *involuting* or *self-inverse* if $f = f^{-1}$ for all x in the domain of f . The function f is defined by

$$f : x \mapsto \frac{2x+p}{x-2} \text{ for } x \in \mathbb{R}, x < 2,$$

where p is a positive constant.

- (i) Show that f is an *involuting* function. [2]
 (ii) By finding $f^3(x)$ and $f^4(x)$, deduce $f^{2021}(x)$. [3]

For the rest of the question, let $p = 2$.

Another function g is defined by $g : x \mapsto \ln(4-x), x < 3$.

Only one of the composite functions fg and gf exists.

- (iii) Give the rule of the one that exists and explain why the other does not exist. [4]
 (iv) Hence find the exact range of the composite function that exists. [2]
4. The plane p has equation $3x - 4y - z = -7$ and the line l has equation $\frac{x-9}{a} = \frac{y+5}{b} = z-2$, where a and b are non-zero constants. The point A with position vector $9\mathbf{i} - 5\mathbf{j} + 2\mathbf{k}$ lies on l .
- (i) Determine a relationship between a and b if l is parallel to p . [3]
 (ii) In the case where $a = 2$ and $b = -2$,
- (a) find the position vector of the point of intersection of l and p , [2]
 (b) find the position vector of the foot of perpendicular from A to p and hence find the shortest distance from A to p . [5]
 (c) find the acute angle between l and p using the results in part (b). [2]

Section B: Statistics [60 marks]

5. (a) In how many ways can 3 letters from the word “HAPPY” be arranged in a row if at least one of the letters is P ? [3]
 (b) A photographer is arranging 4 male and 5 female teachers for a photo shoot. Find the number of ways the photographer can arrange the teachers side by side in a straight line if
- (i) one particular male teacher insists on standing in the middle, [1]
 (ii) the female teachers are arranged in the order from the shortest on the left to the tallest on the right, and no two female teachers are standing next to each other, (You may assume that the heights of the female teachers are all different.) [2]
 (iii) the male teachers are standing next to each other and so are the female teachers. [2]

6. A set of 36 cards is made up of cards chosen from a few packs of ordinary playing cards. The breakdown of the number of cards in each suit and denomination is given in the following table.

Suit \ Denomination	Spade	Heart	Diamond	Club
Ace	1	2	2	1
King	2	2	2	3
Queen	2	4	4	3
Jack	1	3	2	2

For example, there are 2 Aces of Diamond and 3 Jacks of Heart in the set of 36 cards.

- (i) This set of 36 cards is put into a bag and 1 card is selected at random.
- (a) Find the probability that the card is either an Ace or a Jack but not a Club. [1]
- (b) Find the probability that the card is neither a Jack nor a Heart. [1]
- (ii) The cards are all placed back into the bag and 2 cards are selected at random.
- (a) Find the probability that both cards are Aces given that neither of the cards is Spade. [2]
- (b) Find the probability that the 2 cards include exactly one Queen and exactly one Diamond. [3]
7. An infectious virus called Zeevus hit an Asian country Atopia. It was found from data corroborated with other countries in Asia that the virus had a *mean incubation period* of 10.8 days.

The incubation period is defined as the time from exposure to a virus to the development of symptoms. For highly infectious diseases, authorities are guided by the mean incubation period to quarantine suspected infected patients in order to minimise transmission.

A city of Atopia reported that the incubation period had been found to be as long as 20 days in a few cases and this had worried the authorities. A medical researcher was tasked to collect a random sample of 100 patients. The incubation period, x days, are summarised as follows:

$$\sum (x - 12) = 39, \sum (x - 12)^2 = 2716.88.$$

- (i) Calculate the unbiased estimates of the population mean and variance of the incubation period. [2]
- (ii) Explain whether the researcher should carry out a 1-tail or 2-tail test, stating the hypotheses for the tests, and defining any symbols you use. [2]
- (iii) Carry out the test, at 1% level of significance for the researcher. [3]

A few months later, the authorities suspected that a new strain of the same virus might have developed which affected the incubation period. They assigned the same researcher to find out if there was a change in the mean incubation period. Data from 100 randomly selected patients from across the country were collected for the test.

- (iv) Determine the set of values of the sample mean to show that the incubation period of the new strain of the virus is different from the original strain at the 1% level of significance. [3]

8. A company has 50 workers. It is known that, on average, the probability that any particular worker is on sick leave on a given day is p . Let W be the random variable of the number of workers on sick leave on a given day.

- (i) State, in the context of the question, two assumptions needed to model W by a binomial distribution. Explain why one of the assumptions may not be valid in this context. [3]

Assume that the two assumptions in part (i) do in fact hold. A day is selected at random and the probability that at most 2 workers are on sick leave is 0.54.

- (ii) Write down an equation satisfied by p . Hence find the value of p . [3]

Assuming there are 60 working days in 2 months,

- (iii) find the expected number of days in 2 months for which at most 2 workers are on sick leave. [2]
 (iv) find the probability that the mean number of workers on sick leave per day over 2 months is at least 2. [3]

9. A local radio station organises a call-in competition, in which listeners are given the names of 7 local celebrities and they are told that 4 of them are in the studio. Listeners will then call in and give the names of the 4 celebrities they think are in the studio. The random variable X is the number of correct guesses made by a randomly selected listener.

- (i) Show that $P(X = 1) = \frac{4}{35}$. [1]
 (ii) Find the probability distribution of X . [3]
 (iii) Find $E(X)$ and show that $\text{Var}(X) = \frac{24}{49}$. [2]

On any day, 3 listeners will be selected to participate in this competition. The radio station will only reveal which 4 celebrities are in the studio after all 3 listeners have made their guesses.

- (iv) Find the probability that the total number of correct guesses from the 3 listeners is 6. State any assumption you have made in your calculation. [4]
 (v) Find the probability that all 3 listeners make the same guesses. [2]

10. In this question you should state the parameters of any distributions that you use.

A brand of mobile phone launches a new model. Each phone is made up of two key parts – phone body and removable battery. The masses in grams of the phone bodies have the distribution $N(165, 15^2)$.

- (i) Sketch the distribution for masses of the phone bodies within three standard deviations of the mean. [2]
- (ii) Find the probability that the difference between 2 randomly chosen phone bodies is less than 0.5 grams. [3]

It is given further that the mass of a battery is modelled as 18% of the mass of a phone body.

- (iii) According to ergonomics standard, the ideal mass of a phone, **with the battery installed**, is between 140 and 170 grams. Find the probability that a randomly chosen phone is of ideal mass. [3]

A standard travel charger is also enclosed with the phone and it weighs q grams.

- (iv) The probability that the total mass of a phone with a standard travel charger is less than 0.247 kg is 0.85. Find the mass of the travel charger. [4]

END OF PAPER